

Final Project for SW Engineering Class CSC 648-848 Spring 2025

Team 2

Claudia Wormley (Team Lead)
Nathan Donat-Filliod (GitHub Lead)
Daniel Cervantes (Back-End Lead)
Davis Rosenstein (Front-End Lead)
Fatimah Abdolcader
cwormley@sfsu.edu

Buy-Sell WWW app - GatorMarket
<http://44.201.159.31/>

Demo: 5-21

Product summary

Name: Gator Market

Product Functions:

1. All users are able to view recent products
2. All users are able to filter products using categories
3. All users are able to search for products using keywords
4. All users are able to search for books by searching the course number of the class it is for
5. All users are able to view product title, description, image, and price
6. All users are able to create an account only with a valid sfsu email.
7. All users are able to sign in once they've created an account
8. Logged in users are able to message a product seller.
9. Logged in users are able to specify meetup location based on our designated meet up spots
10. Logged in users are able to create a new product listing with a title, description, price, category, and image.
11. Logged in users are able to view their listings and incoming messages from their dashboard.
12. Logged in users are able to delete their listings
13. Admin users are able to review and approve posts before the product is public.
14. Admin users are able to delete user accounts and postings

GatorMarket Specific Features:

GatorMarket is a marketplace designed exclusively for the San Francisco State University community, allowing students, faculty, and staff to buy and sell items like textbooks, electronics, and furniture. What makes it unique is its exclusive access via SFSU email, a class-specific search for course materials, and safe on-campus pickup locations—all tailored to the campus experience. Created by SFSU computer science students, GatorMarket emphasizes simplicity, security, and community connection, standing apart from platforms like Craigslist or Facebook by focusing only on what SFSU users need.

URL: <http://44.201.159.31/>

Product Screenshots

SFSU Software Engineering Project CSC 648-848, Spring 2025. For Demonstration Only



Post Item

Dashboard

All Categories

Search...

Search

About

Map

Sign In

Register

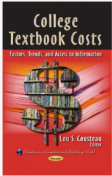
Welcome to Gator Market!

A community marketplace for SFSU students to buy and sell textbooks, supplies, and more.

Find what you need or list your items for other students - all in one convenient place.

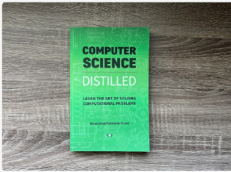
Recent Posts

Standard TextBook



This is a standard text book. Used. 44.201.159.31/item/111

Intro Java



Intro to Java textbook, recommended reading for CSC 101.

Charger



Laptop Charger. Compatible with HP devices. Fine condition.

Bike



Well used. Some scrapes and bumps but rides well. Tires need a little love.

SFSU Software Engineering Project CSC 648-848, Spring 2025. For Demonstration Only



Post Item

Dashboard

Class Books

book

Search

About

Map

Log Out

Welcome to Gator Market!

A community marketplace for SFSU students to buy and sell textbooks, supplies, and more.

Find what you need or list your items for other students - all in one convenient place.

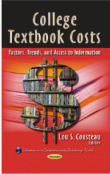
Search Results

Intro Java



Intro to Java textbook, recommended reading for CSC 101.

Standard TextBook



This is a standard text book. Used. not required for any classes

SFSU Software Engineering Project CSC 648-848, Spring 2025. For Demonstration Only

GATOR MARKET

Post Item

Dashboard

All Categories ▾

Search...

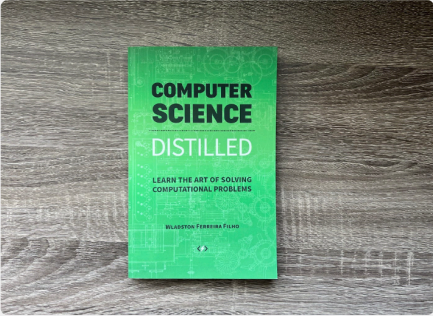
Search

About

Map

Sign In

Register



Intro Java

\$10

Category: Class Books

Course: CSC 101

Description:

Intro to Java textbook, recommended reading for CSC 101.

Contact Seller

SFSU Software Engineering Project CSC 648-848, Spring 2025. For Demonstration Only

GATOR MARKET

Post Item

Dashboard

All Categories ▾

Search...

Search

About

Map

Sign In

Register

Intro Java



\$10

Category: Class Books

Course: CSC 101

Enter Contact Info:

Enter Phone or email for seller to contact you

Enter Your Message to the Seller:

Please include an on-campus meeting spot from our designated [Safe Meetup Locations](#)

Type your message here

Cancel

Sign In to Post



SFSU Software Engineering Project CSC 648-848, Spring 2025. For Demonstration Only

Post Item

Dashboard

All Categories ▾

Search...

Search

About

Map

Sign In

Register

Register

Email:

Password:

Confirm Password:

☐ I accept the [Terms and Conditions](#)

Register

Already have an account? [Sign In](#)



SFSU Software Engineering Project CSC 648-848, Spring 2025. For Demonstration Only

Post Item

Dashboard

All Categories ▾

Search...

Search

About

Map

Sign In

Register

Sign In

Email:

Password:

[Forgot Password?](#)

Sign In

Don't have an account? [Register](#)

GATOR MARKET

Post ItemDashboard

All CategoriesSearch...Search

AboutMapLog Out

Post an Item

* All fields mandatory

Title:

Price:

Category:

Select a category

Description:

Enter a description of the item

Image:

Choose FileNo file chosen

Post Item

Posts will be approved within 24 hours

SFSU Software Engineering Project CSC 648-848, Spring 2025. For Demonstration Only

GATOR MARKET

Post ItemDashboard

All CategoriesSearch...Search

AboutMapLog Out

Dashboard

Welcome back test! Here's a summary of your activity..

Messages: 1

Total Listings: 4

Post an Item

MessagesListings

Messages

Item: Charger

From: new

Contact: example@gmail.com

Date: 5/19/2025



Hello! I would like to buy your charger. Would you be able to meet on campus outside of the student center? Feel free to message me!

Dashboard

Welcome back test! Here's a summary of your activity..

Messages: 1

Total Listings: 4

[Post an Item](#)

Messages

Listings

Your Listings

Date: 5/19/2025

Delete

Intro Java



Price: \$10

Status: Approved

Date: 5/19/2025

Delete

Charger



Price: \$10

Status: Approved

Date: 5/19/2025

Delete

Bike



Price: \$40

Status: Approved

SFSU Software Engineering Project CSC 648-848, Spring 2025. For Demonstration Only



[Post Item](#)

[Dashboard](#)

All Categories

Search...

Search

[About](#)

[Map](#)

[Log Out](#)

Meet Team 02!

We are a group of students from SFSU, working on a software engineering project for our CSC 648-848 class.

Team Members



Cj



Daniel



Nathan



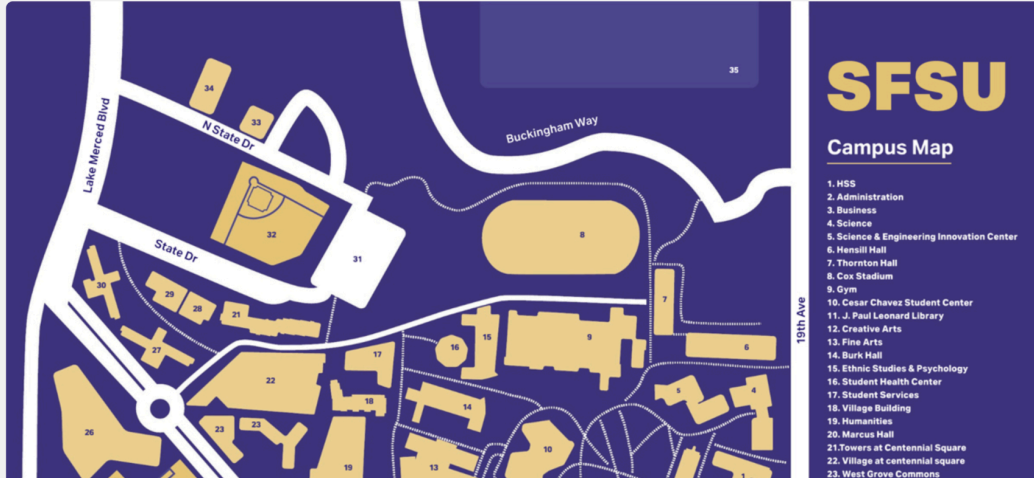
Fatimah



Davis

[Post Item](#)[Dashboard](#)[All Categories](#)[Search](#)[About](#)[Map](#)[Log Out](#)

Safe Meet-up Locations on Campus



Database Organization

```
[mysql> Describe Product;
```

Field	Type	Null	Key	Default	Extra
product_id	int unsigned	NO	PRI	NULL	auto_increment
user_id	int unsigned	NO	MUL	NULL	
title	varchar(100)	NO		NULL	
description	varchar(2000)	NO		NULL	
category_id	int unsigned	NO	MUL	NULL	DEFAULT_GENERATED
price	float	NO		NULL	
created_at	timestamp	YES		CURRENT_TIMESTAMP	
images	text	NO		NULL	
approved	int	NO		0	
class_name	varchar(255)	YES		NULL	

```
10 rows in set (0.02 sec)
```

```
[mysql> Describe User;
```

Field	Type	Null	Key	Default	Extra
user_id	int unsigned	NO	PRI	NULL	auto_increment
email	varchar(50)	NO		NULL	
password	varchar(50)	NO		NULL	
is_admin	tinyint(1)	NO		NULL	

```
4 rows in set (0.00 sec)
```

```
[mysql> Describe Message;
```

Field	Type	Null	Key	Default	Extra
message_id	int unsigned	NO	PRI	NULL	auto_increment
product_id	int unsigned	NO	MUL	NULL	
user_id	int unsigned	NO	MUL	NULL	
content	varchar(2000)	NO		NULL	DEFAULT_GENERATED
timestamp	timestamp	YES		CURRENT_TIMESTAMP	
user_contact	varchar(100)	NO		NULL	

```
6 rows in set (0.00 sec)
```

```
[mysql> Describe Category;
```

Field	Type	Null	Key	Default	Extra
category_id	int unsigned	NO	PRI	NULL	auto_increment
name	varchar(50)	NO		NULL	

2 rows in set (0.01 sec)

Github organization

Branches:

main

└_ dev

└_ development branch #1 (ex. m3revisions)

└_ development branch #2 (ex. homePage)

Master branch:

The master branch was only accessible to team lead and github lead. This protected the production version of our product by avoiding any unintentional pushes to main.

The screenshot shows a GitHub repository page for 'csc648-sp25-03-team02' (Private). The repository is forked from 'CSC-648-SFSU/csc-648-sfsu-2025-fall-petkovic-csc648-base-repo'. The main branch is selected, showing 4 branches and 0 tags. The repository is 161 commits ahead of the main branch. The file list includes: .github/workflows, Milestones, application, credentials, .DS_Store, .gitignore, LICENSE, README.md, package-lock.json, and package.json. The right sidebar shows repository statistics: 0 watches, 0 forks, and 1 star. It also includes sections for About, Releases, and Packages.

Google analytics

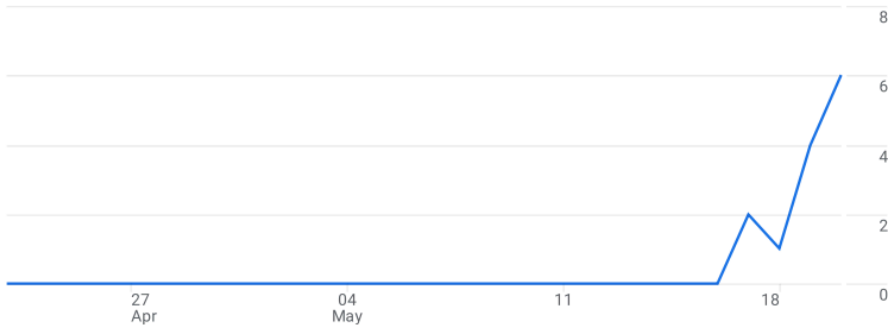


All Users

Add comparison +

Reports snapshot

< **Active users** ? **New users** ? **Average engagement time per active u** 5m 18s > ✓



Active users ▾ by Co



Views by Page title and screen class ✓ ▾

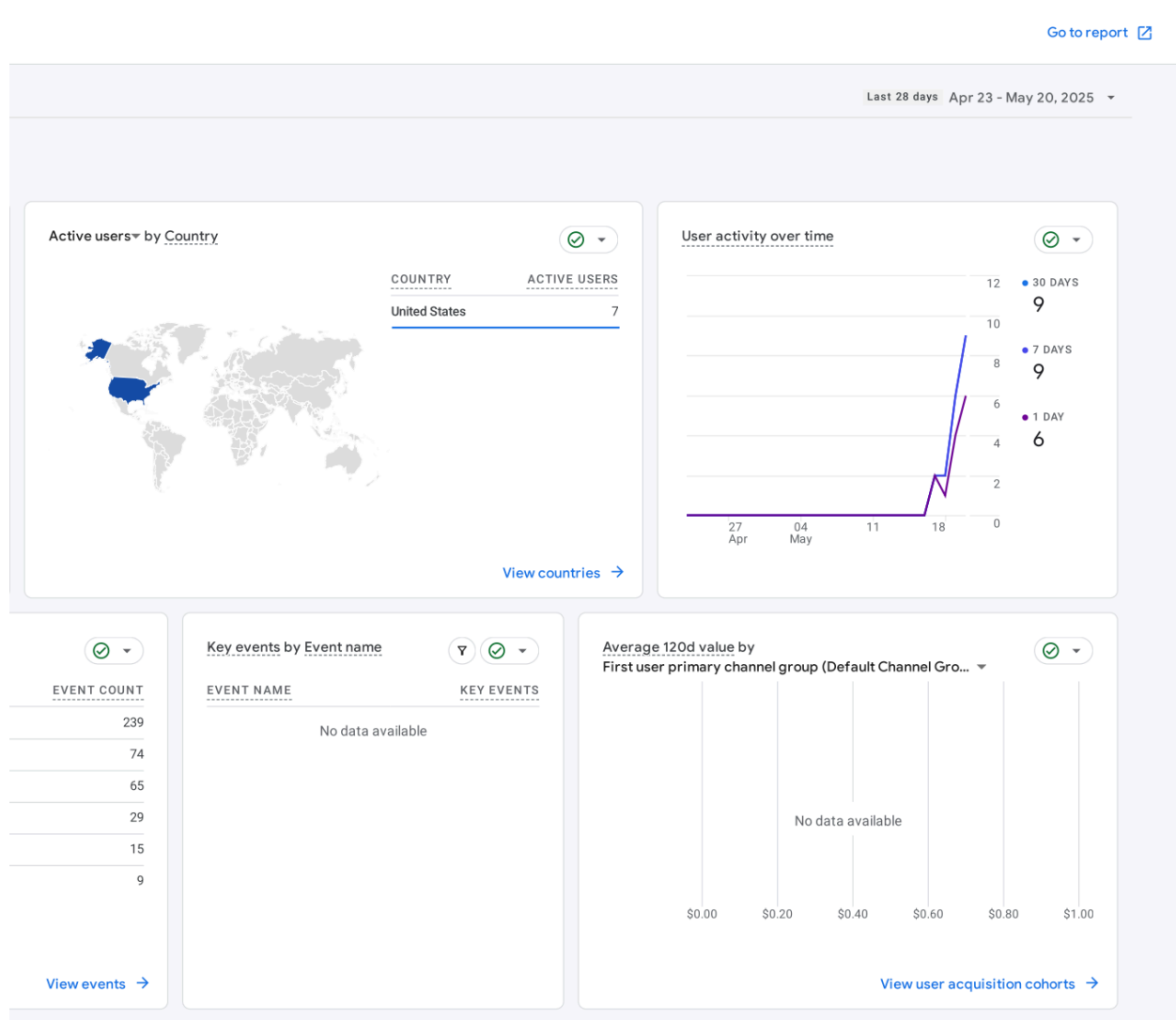
PAGE TITLE AND SCREEN CLASS	VIEWS
CSC 648 Project	239

[View pages and screens →](#)

Event count by Event name ✓ ▾

EVENT NAME	EVENT COUNT
page_view	239
scroll	74
user_engagement	65
form_start	29
session_start	15
first_visit	9

[View events →](#)



Project management

For the majority of production our team relied on discord for assigning tasks, sharing progress and updating the team. While trello was briefly used, it was not well adapted by the team. Trello requires more updating and monitoring and there is no direct line of communication. As team lead I needed a better way of monitoring progress that didn't rely on teammates manually updating the trello page. Discord allowed me to check in directly with my teammates and keep them updated on progress. The great thing about discord is that I can better track who has seen and started on assignments. Trello assumes the team keeps it actively updated.

Team 2

Board

To Do

✓ Implement frontend homepage

16 Apr

F

Set up db access in WorkBench - Meet with CTO

16 Apr

DC

Doing

Implement frontend dashboard

16 Apr

DR

Implement changes to M2 Doc

13 Apr

DC

Implement changes to M2 Doc

13 Apr

F

Done

✓ Implement changes to M2 Doc

13 Apr

DR

✓ Create tables and Example products in SQL

✓ Secure image folder

14 May

✓ Self-check on best practices for security

✓ Implement changes to M2 Doc

13 Apr

1

✓ Implement unit tests and try to automate it with CI/CD

23 Apr

0/5

CSC 648 Team 2

calendar

Events

Server Boosts

Information

current-milestone

calendar

resources

Team Collab

frontend

backend

Text Channels

team-member-schedules

general

help-and-updates

github

ai-log

Voice Channels

Milestone 1

Peer Reviews - Due Friday March 7th

Fatimah review Davis's Use Cases

Davis review Daniel's Data Entities

Daniel review Nathan's Functional Specifications (Don't need to review Non Functional)

Nathan review Fatimah's Personae

Fatimah and Davis - #7 Competitive analysis

Daniel - #8 High-level system architecture

Nathan - #10 Team and roles

Everyone - Include a summary of how you used AI include rating and a sample prompt (edited)

1

12 March 2025

Cj 12/03/2025, 22:10

Milestone 2

Revisions of Milestone 1 - implement the changes prof suggested - Due 3-13

1. List of main data items and entities - **Daniel** Due 3-14

2. Functional Requirements - **Nathan** Due 3-17

3. UI Storyboards - **Fatimah and Davis** (Rough Draft 3-19)(Final Draft 3-21)

4. High level Architecture - **Daniel and Nathan** Due 3-19

5. Milestone 2 Part II: Vertical SW Prototype - **Daniel and Nathan** Due - 2-26

6. AI Log - **ALL** Due 3-22

Team member self assessment



© Claudia Jane Wormley <cwormley@sfsu.edu>

Today at 2:42 PM

To: © Daniel Cervantes; © Davis John Rosenstein; © Fatimah Inayah Abdolcader; © Nathan Pierre Jean Donat Filliod

Hi Team 02,

Below is my self-assessment for my role as team lead.

Team Lead Contributions:

- Assigned individual team member roles for each milestone and development phase.
- Communicated with team regularly, providing and requesting updates.
- Formatted and submitted milestone documents
- Contributed to Milestone documentation
- Provided feedback on team member work
- Implemented Backend API's
- Assisted team members when needed, troubleshooting/resolving errors
- Communicated with CEO to verify milestone expectations and provide updates
- Full-stack development for Demo preparation
- Attended all class and team meetings
- Completed Milestone 5 Document and presented Demo

GitHub commits to Dev: 86

Challenges:

Outside of my roles as team lead, I encountered challenges in the early development phases. It is difficult to see the project as a whole in the beginning so building the SW stack and setting team expectations came with difficulty. It was only after developing the use cases and getting a better idea of our backend set up that the whole picture came together. I was not as experienced in the backend when I started so working through that development was hard but was very helpful in the long run.

GenAI:

ChatGPT and copilot were very helpful in documentation and production. I was able to use ChatGPT to assist in reviewing milestone writeups and creating product descriptions. Copilot was incredibly helpful as a coding assistant and provided great header and inline comments. While they both have their limitations and need to be used in a limited scope, they can be a great assistance in formatting and general programming assistance.

SE management and processes:

This class gave me a strong understanding of all the factors outside of coding that go into creating a product. Understanding the roles of piece of documentation and each stage in the process helped me be able to utilize them more effectively. Going forward I will go into the project with a better understanding of how everything works together and how to implement a strong foundation by developing project scope and expectations early on

Overall feedback from team lead:

As team lead, I was responsible for the overall success of the project and team. To do so I needed to delegate responsibilities, communicate regularly with my team, and encourage productivity. I encountered challenges in balancing my responsibilities and delegation. I had to learn to trust my team to do what they needed to do, and I was very lucky to have a team full of talented programmers that met expectations on every milestone. As the semester went on, I learned to rely on them and trust that they would do their part. This made it easier for me to focus on the overall picture. Communication was also a struggle at times because we had limited time to work in person. I learned to prepare in advance so that the time we had in class was well spent. This along with regular communication through discord helped keep the team on track.

ChatGPT was very helpful in formatting the milestone documents and delegating tasks in the early phases. As this was my first time in a team lead role I was able to use GenAI to help in developing team expectations and explaining any parts of the assignments I didn't fully understand. This helped me become more confident over the course of the semester and rely on it less in later milestones. Copilot was also helpful in code reviews. As I was reviewing and editing both front-end and back-end development copilot assisted in highlighting sections that need attention and explaining the new implementations.

This class taught me so much that I will be bringing into future projects and jobs. This process taught me how vital management and scheduling are to larger projects. Being team lead also helped me see the project as a whole and saw how all parts fit together. If I were to do this all again, I would pay more attention to the foundational phases. Implementing team expectations and a solid SW stack in the very beginning to help the project run more smoothly.



📧 Daniel Cervantes <dcervantes2@sfsu.edu>

Today at 1:54 PM

To: 📧 Claudia Jane Wormley; 📧 Nathan Pierre Jean Donat Filliod; 📧 Davis John Rosenstein; 📧 Fatimah Inayah Abdolcader

Hello team,

I was the back-end lead, I contributed to the project by setting up the server and database. My job was to get the server started, fix any issues with the server and database, and store important login information. I also created the AWS account, which allowed us to host our server there.

39 submissions were made to the GitHub team by me.

Some of the main challenges I faced were setting up the database and trying to connect it locally. I had trouble with time management as well because I had multiple other group projects, so balancing it all was tough.

Using GenAI was useful because it made the process time-efficient and also made things a lot easier, as I was able to provide a prompt, and it generated a solution. All I had to do was review, make changes, and implement.

One thing I would do a lot better would be to communicate more often and also be more time efficient in terms of being organized with other projects. I would work on being more efficient also in my code, and also get a better understanding of the assignment in terms of what is needed and not doing more/extra work.

Best to all,
Daniel Cervantes



📧 Davis John Rosenstein <drosenstein@sfsu.edu>

Today at 2:44 PM

To: 📧 Claudia Jane Wormley; 📧 Fatimah Inayah Abdolcader; 📧 Nathan Pierre Jean Donat Filliod; 📧 Daniel Cervantes

Hi Team,

- My contributions - Front end lead:
 - Worked on the UI storyboards, use cases, and test plan in milestones
 - Created the main about page
 - Created and designed the dashboard
 - Created and designed the post item page
 - Created and designed the nav bar
 - Created and designed signin and register pages
 - Worked on general CSS for the website
- I only had 9 commits, low commit count because I would make a lot of changes or finish a page before making a commit, then make all the fixes I needed in another commit.
- One of the main challenges I faced was merging conflicts. There were times when some of the changes I made got accidentally deleted because of merging conflicts so I had to re add them. There were also some changes I wanted to make but couldn't find where they were because the code wasn't in the file I thought it would be in.
- Gen AI was useful in helping me complete my part of the project. Copilot was useful in getting started with adding features to pages I wasn't sure how to implement or get started with. It was also helpful getting started on some of the written portions in the milestone documents. I find the use of GenAI helpful because I often struggle with where to start so just having something to start from even if it is not the most useful helps greatly with how fast I can finish an assignment.
- One thing I would do differently is try to communicate more with my team and the team lead to be better organized when working on the project. I would also try to make more commits and comments in my code to make it easier for other to go through if they need to make any changes or check how something works.



📧 Fatimah Inayah Abdolcader <fabdolcader@sfsu.edu>

Today at 1:42 PM

To: 📧 Claudia Jane Wormley; 📧 Nathan Pierre Jean Donat Filliod; 📧 Daniel Cervantes; 📧 Davis John Rosenstein

Hi team,

As part of the front-end team, I contributed in the following ways:

- Designed the initial wireframes and page layouts for the "About Me" pages using Figma.
- Helped create and refine the scenario designs in Figma. Developed the front-end implementation of several core pages, including:
 - Home page
 - Listing page
 - Map page
 - Message Seller page
- All branches total commits: 30
- The main challenges I encountered when in a team project was merging code with a whole bunch of changes. It could seem confusing when adjusting code that has been tampered with. For the most part I was able to figure out how to implement both changes or either or. Overall, it could be fun fixing the conflicts being changed.
- My experience with GenAI has been a journey as I realized how to appropriately use this as a tool. I found GitHub Copilot very helpful when it came to polishing final edits. The structure of my code seemed better when gaining feedback from AI. I thoroughly enjoyed the help I would be able to gain when confused about a certain task or what I could improve on. I understood that not all feedback from AI was correct, so I made sure to read thoroughly.

- What I could do better with being in a team was more communication with my front-end lead so the tasks could run smoother. My availability was low so that also played a major factor. I could do better logically with how I went to structure my code. I feel like I did not take into account more high level logic that could have been incorporated. I could overall be better with my time.
- Wednesdays and Fridays were the only days I did not work for the most part and I found it challenging to take my time on tasks as I was working after school everyday.

Best,
Fatimah



ⓧ Nathan Pierre Jean Donat Filliod <ndonatfilliod@sfsu.edu>

Today at 3:50 PM

To: ⓧ Claudia Jane Wormley; ⓧ Daniel Cervantes; ⓧ Davis John Rosenstein; ⓧ Fatimah Inayah Abdolcader

Hi Team 02,

Github Master Contributions:

- Review of pull requests
- Configuration of a CI
- Storage of variables in .env
- Database setup
- Creation of a secured folder in server for images

GitHub commits to Dev branch: 30

Main challenges encountered:

- Conflicting changes in the same files/lines from different branches.

Using of GenAI:

I used AI to help with my project, but it caused a lot of errors that I had to fix manually, which made the process frustrating and time-consuming.

However, I found it useful for drafting ideas and writing out some key milestones, it helped me organize my thoughts even if I couldn't rely on it for technical accuracy.

SE management and processes:

Next time, I'll set up CI/CD earlier in the project to catch issues automatically and ensure everything runs smoothly from the start. I'll also be more careful with commit messages to keep the history clean and understandable, and I'll take code reviews more seriously to maintain quality and avoid unnecessary bugs.